

A Fixed-Income Analytics Engine:

Mathematical Foundations, Numerical Verification, Yield-Curve
Modelling and Interest-Rate Risk

Fixed-rate bond valuation and the Nelson–Siegel–Svensson term
structure

A computational finance dissertation

Educational computational study.

No institutional affiliation or submission is asserted.

Study status and declaration note

This dissertation studies an existing educational computational framework. Its inputs are invented offline observations; no institutional affiliation, submission or empirical market validity is asserted. The independent internal verification record remains available as `paper/verification_report.tex`.

Abstract

Fixed-rate bond valuation requires consistency between payment timing, discounting and the interest-rate variable being changed. This dissertation examines calculations for regular bonds, smooth term-structure representation and deterministic curve risk. Nominal coupon-compounded valuation is distinguished from discounting with continuously compounded zero rates. Analytical differentiation establishes duration and convexity; a six-parameter Nelson–Siegel–Svensson function represents illustrative maturity-rate observations. Independent cash-flow calculations, high-precision exponential references, finite differences and a partition-of-unity proof assess the implementation. All 234 deterministic tests pass. The illustrative 2030 bond has a clean price of 97.876838 per 100 face value and modified duration of 4.231518 years. The curve-fit root mean squared error is 0.782762 basis points. For the longer risk bond, parallel sensitivity is 0.1697264000 currency units per one-basis-point bump; summed key-rate sensitivity differs by approximately minus 2.93×10^{-8} currency units, consistent with a cubic correction. The principal limitation is that fitted illustrative rates are assigned a zero-rate interpretation without inference from instrument prices. The evidence supports mathematical consistency under explicit conventions; empirical market validity remains untested.

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Notation and abbreviations

The notation distinguishes nominal yield-to-maturity valuation from explicit zero-curve discounting. Rates are decimals unless a percentage or basis-point display is explicitly stated. Currency amounts refer to the supplied face value. Tables 1 to 3 collect symbols, abbreviations and terminology; chapter definitions introduce their financial meaning.

Table 1: Notation, units and first-use locations

Symbol	Meaning	Units and convention	First use
F	Face and redemption amount	Currency; default 100	3.1
c	Annual coupon rate	Decimal	3.1
m	Coupon frequency	Payments per year; 1 or 2	3.1
C	Coupon per payment	Currency	3.1
y_{YTM}	Yield to maturity	Nominal annual decimal; compounded m times	3.1
B	$1 + y_{\text{YTM}}/m$, accumulation per period	Dimensionless; positive	3.1
d_s	Settlement date	Calendar date	3.1
d_-, d_+	Previous/on-settlement and next coupon dates	Calendar dates	3.1
d_i	Date of future payment i	Calendar date	3.1
days	Elapsed calendar-day count	Days	3.1
α	Accrued fraction of coupon period	Dimensionless	3.1
N, i	Future payment count and payment index	Integer; $1 \leq i \leq N$	3.1
q_i	Fractional coupon periods to payment	Dimensionless period count	3.1
T_i	q_i/m , YTM payment time	Coupon-based years	3.1
CF_i	Coupon plus redemption if final payment	Currency	3.1
PV_i	YTM-discounted cash-flow present value	Currency	3.1
AI	Accrued interest	Currency	3.1

Symbol	Meaning	Units and convention	First use
$P_{\text{dirty}}, P_{\text{clean}}$	Prices including/excluding accrued interest	Currency	3.1
$P(y_{\text{YTM}})$	Abbreviation for dirty YTM price	Currency	3.1
P', P''	First/second YTM derivatives	Currency per decimal yield / yield squared	3.2, 3.2
D_{Mac}	Macaulay duration	Years, using T_i	3.2
D_{Mod}	Modified duration, $-P'/P$	Conventional years; annual decimal YTM	3.2
$\mathcal{C}$	Convexity, P''/P	Conventional years squared	3.2
h	Symmetric perturbation magnitude	Decimal rate; 10^{-4} for DV01	3.2
δy	Change in annual nominal YTM	Decimal yield	3.2
DV01_{YTM}	Central YTM repricing sensitivity	Currency for 1 bp	3.2
$O(h^p)$	Local asymptotic remainder order	Units of expression it completes	3.2, A.3
T	Generic curve maturity	Years; positive	2.1
$\theta, \hat{\theta}$	NSS parameter vector; selected fit	Mixed rate/year coordinates	4, 4.1
$\beta_0, \beta_1, \beta_2, \beta_3$	NSS level/loading coefficients	Decimal rate	4
τ, τ_1, τ_2	Positive NSS decay scales	Years, not payment times	4
x	Loading ratio T/τ ; generic exponential argument in Appendix A	Dimensionless	4
$L_1(T; \tau), L_2(T; \tau)$	NSS loading functions	Dimensionless	4
$r_{\text{NSS}}(T; \theta)$	Generic fitted NSS rate	Decimal; caller's convention	4
n, ℓ	Observation count and index	Integers; $1 \leq \ell \leq n$	4.1
u_ℓ, z_ℓ	Observation maturity and rate	Years; decimal rate	4.1
e_ℓ	Observed minus fitted residual	Decimal rate	4.1
$S(\theta)$	Sum of squared rate residuals	Squared decimal rate	4.1
RMSE, RMSE _{bp}	Root mean squared rate error	Decimal rate; basis points	4.1
t_i	days($d_i - d_s$)/365	Actual/365 years	5
$r(T)$	Explicit zero rate	Continuous annual decimal	2.1
$DF(T)$	Zero-curve discount factor	Dimensionless	2.1
V_i	Zero-discounted cash-flow present value	Currency	5

Symbol	Meaning	Units and convention	First use
$P_{\text{zero}}[r]$	Dirty price as a function of zero curve	Currency	5
$s(T), \delta r$	Curve shock; constant parallel amplitude	Decimal zero-rate change	5
ΔP_s	Shocked minus base price (scenario P&L)	Currency	5
J, j, k_j	Key count, index and tenor	Integers; tenor in years	5.1
$b_j(T)$	Key-rate basis weight	Dimensionless	5.1
KRDV01_j	Central key-rate sensitivity	Currency for 1 bp basis move	5.1
$\text{DV01}_{\parallel}$	Central parallel zero-rate sensitivity	Currency for 1 bp	5.1
Δ_{KR}	Sum of key DV01 minus parallel DV01	Currency	5.1
$\sinh, \cosh$	Hyperbolic sine and cosine	Dimensionless arguments	A.6
$P^{(p)}, p$	p -th YTM derivative; integer order/power	Derivative units; positive integer	A.3
$r_s(T)$	Shocked zero rate, $r(T) + s(T)$	Continuous annual decimal	5
u_j	Amplitude of key j 's shock	Decimal rate	A.6
$r_j^{\pm}, P_j^{\pm}$	Key-bumped zero curves and their dirty prices	Decimal rates; currency	A.6
r'_{NSS}	NSS derivative with respect to maturity	Decimal rate per year	A.4
Δ_3	Cubic term in key/parallel reconciliation	Currency	A.6

The cash-flow index i and observation index ℓ are distinct. Coupon-based T_i and Actual/365 t_i retain separate timing conventions.

Abbreviations

Table 2: Abbreviations

Abbreviation	Expansion
CMT	Constant maturity (Treasury par-style yield quotation)
AI	Accrued interest
bp	Basis point; 10^{-4} decimal rate
BIS	Bank for International Settlements
CLI	Command-line interface; existing illustrative commands only
DV01	Currency price sensitivity for a one-basis-point rate move

IMF	International Monetary Fund
KRDV01	Key-rate DV01
NBER	National Bureau of Economic Research
NSS	Nelson–Siegel–Svensson
P&L	Profit and loss; here instantaneous price change only
PV	Present value
RMSE	Root mean squared error
SSE	Sum of squared errors; $S(\theta)$ here
YTM	Yield to maturity

Terminology

These definitions supplement the concise explanations in the chapters.

Table 3: Terminology, readable definitions and first-use locations

Term	Definition	First use
Bond	A debt contract specifying payments from an issuer to its holder.	2
Fixed-rate bond	A bond whose contractual coupon rate is fixed; this study also holds promised cash flows fixed.	2
Face or par value	The contractual reference amount used to calculate coupons and, here, repaid at maturity.	2
Principal	The contractual capital amount, distinct from interest payments.	2
Redemption payment	The repayment of principal at the bond's maturity; here it equals face value.	2
Coupon	A contractual interest payment in currency; it is distinct from the coupon rate.	2
Coupon rate	The annual contractual interest amount divided by face value, expressed internally as a decimal.	2
Coupon frequency	The number of scheduled coupon payments per year; one or two in this study.	2
Settlement date	The valuation and transfer date from which entitlement to future payments is determined under the stated convention.	2
Maturity date	The contractual date of final principal repayment.	2
Regular coupon schedule	A sequence of payments at the prescribed frequency without irregular first or final periods.	2
Accrued interest	The coupon amount attributed to the elapsed part of a coupon period under a specified accrual rule.	2
Dirty price	The value including accrued interest; here the discounted value of the strictly future cash flows.	2
Clean price	The dirty price less accrued interest under the selected quotation convention.	2
Yield to maturity	The single discount yield that equates promised cash-flow present values to dirty price under specified timing and compounding.	2.1
Nominal compounding	Quotation of an annual rate whose periodic rate is the annual rate divided by compounding frequency; fractional periods require an explicit extension.	2

Term	Definition	First use
Continuous compounding	The convention in which accumulation at a constant annual rate over a given horizon is exponential in rate times time.	2
Discount factor	The present value of one unit payable at a specified future time under a stated discounting rule.	2
Present value	The value at the valuation date assigned to a future payment by discounting it.	2
Time value of money	The dependence of the comparison between present and future amounts on their payment dates and the accumulation rule.	2
Basis point	One hundredth of one percentage point: a decimal-rate change of 0.0001.	2.1
Percentage point	An absolute difference between percentage quotations; one percentage point is 0.01 in decimal rates.	2.1
Par, premium and discount	Par means price equal to face; premium and discount mean above and below face, with the price basis stated.	3.1
Duration	A family of measures relating payment timing or local price response to a specified interest-rate variable.	3.2
Macauley duration	The present-value-weighted mean time to payment, using the stated YTM discounting and time convention.	3.2
Modified duration	The negative first price derivative divided by dirty price, with respect to annual decimal YTM.	3.2
Convexity	The second derivative of dirty price divided by that price, with respect to annual decimal YTM.	3.2
DV01	A currency-valued sensitivity for a one-basis-point move; this project defines it by half the down-minus-up repriced difference.	3.2
Taylor approximation	A local polynomial representation built from a function's derivatives; omitted terms determine truncation error.	3.2
Term structure of interest rates	The dependence of interest rates on maturity at a fixed valuation date under common rate and instrument conventions.	2.1
Yield curve	A representation of rates against maturity; its interpretation requires stating which kind of rate is plotted.	2.1
Zero-coupon bond	A bond making no interim coupon payments and a single redemption payment.	2.1
Spot rate	A rate applicable from valuation to one maturity for a single payment, under a specified compounding convention.	2.1
Zero-coupon rate	The single-horizon discount rate associated with a zero-coupon payment; used synonymously with spot rate here.	2.1
Par yield	The annual coupon rate that makes a specified bond worth face value under given discount factors, conventionally at coupon-date settlement without accrual.	2.1

Term	Definition	First use
Parameter	A numerical coefficient or scale specifying a member of a mathematical model family.	4
Calibration	Selection of model parameters to match supplied observations according to an explicit objective.	4.1
Residual	The difference between an observed and fitted value; the sign convention here is observed minus fitted.	4.1
Least squares	A fitting criterion that minimises the sum of squared residuals.	4.1
Nonlinear optimisation	Numerical search for parameters minimising an objective that is nonlinear in the relevant parameters.	4.1
Local minimum	A feasible point whose objective is no larger than at sufficiently nearby feasible points; it need not be globally best.	4.1
RMSE	The square root of the average squared residual; its units match the fitted variable.	4.1
Nelson–Siegel model	A parametric curve with level, slope-related and curvature loadings governed by a positive decay scale.	4
Svensson extension	The addition of a second curvature loading with its own coefficient and decay scale to Nelson–Siegel.	4
Factor loading	A function of maturity multiplying a model coefficient; a loading need not be a directly observed economic variable.	4
Asymptotic limit	A value approached by a function as its argument tends to a boundary such as zero or infinity.	4
Identification	The ability of observations and model restrictions to distinguish parameter values; weak identification means different values can fit very similarly.	4.1
Bootstrap	Sequential inference of discount factors from instrument prices and cash flows under stated conventions; not implemented here.	2.1
Parallel shift	The same additive rate change at every maturity.	5
Full repricing	Recomputing all discounted cash-flow values under a shocked rate or curve, instead of using a local linear approximation.	5
Scenario P&L	The shocked price minus base price, holding settlement and cash flows fixed; it is not a realised investment return.	5
Steepener	A specified curve change increasing the long-minus-short rate difference; this study states the anchor shifts explicitly.	5
Flattener	A specified curve change decreasing the long-minus-short rate difference; it need not make the whole curve flat.	5
Piecewise-linear interpolation	Joining neighbouring prescribed values by straight-line segments; this study holds endpoint values constant outside the anchor range.	5
Key rate	A rate at a selected maturity used to parameterise a local perturbation of the curve.	5.1
Key-rate basis	A set of maturity-dependent weights specifying how a perturbation at each chosen key affects the curve.	5.1

Term	Definition	First use
Partition of unity	A family of weights whose sum is one at every maturity in the specified domain.	5.1
Key-rate DV01	A currency sensitivity obtained by central full repricing under plus and minus one basis point times one key-rate basis.	5.1
Finite difference	An approximation to a derivative constructed from function values at distinct nearby arguments.	6
Central difference	A finite difference using symmetric positive and negative perturbations around the evaluation point.	6
Truncation error	The error due to omitting terms in an approximation, distinct from floating-point rounding.	6
Numerical tolerance	A specified threshold for accepting a comparison or stopping an algorithm; it must be interpreted in its own units and context.	6
Floating-point rounding	Representing a numerical result by a nearby value from the finite set available to the machine.	4
Cancellation	Loss of relative accuracy when nearly equal rounded quantities are subtracted.	4
Conditioning	Sensitivity of the exact solution of a mathematical problem to small input perturbations.	6
Numerical stability	The extent to which an algorithm controls the effects of rounding and related computational perturbations.	4
Deterministic test	A check with fixed inputs and defined expected properties, designed to be repeatable in a specified environment.	6
Actual/365 timing	Actual elapsed calendar days, including leap days, divided by 365; the denominator does not change in leap years.	5
Model risk	The risk of misleading conclusions caused by an unsuitable model, interpretation or set of assumptions.	9
Tenor	A stated time horizon or maturity, measured in years here.	2.1

Chapter 1

Introduction

Bonds finance borrowing through promised payments. Their valuation depends on when those payments occur and how future amounts are discounted (Tuckman and Serrat 2011). A single discount yield cannot generally represent rates that vary by maturity; a yield curve records rates against maturity under a stated convention (Tuckman and Serrat 2011). Price levels alone do not establish rate sensitivity.

This dissertation examines fixed-rate valuation, Nelson–Siegel–Svensson modelling and deterministic interest-rate risk through four questions:

1. How do explicit conventions determine bond prices and yield sensitivities?
2. How can a smooth parametric curve be fitted reliably?
3. How can parallel, shaped and key-rate exposures be represented consistently?
4. What independent evidence supports computational correctness?

The scope comprises regular option-free bonds, whose contracts contain no rights to alter payments, and illustrative offline rates. Credit events, forecasting, trading and backtesting are excluded. The contribution is independent verification of established mathematics, not new financial theory.

Chapters 2 to 5 develop the financial argument; chapters 6 to 8 present computation and evidence. Chapters 9 and 10 delimit the conclusions; appendices A and B supply derivations and reproduction instructions.

Chapter 2

Fixed-income foundations

Money available today can earn interest, so payment dates affect economic value. Present value expresses a future payment in today's units; its discount factor is today's value of one future unit. Discrete compounding credits interest periodically; a nominal annual rate is divided by the compounding frequency. Continuous compounding uses exponential accumulation (Tuckman and Serrat 2011).

A fixed-rate bond is a debt contract paying fixed interest coupons and returning principal at maturity. Face, or par, value is the principal reference amount; redemption repays it. The annual coupon rate determines annual interest, divided into coupon amounts according to payment frequency (Tuckman and Serrat 2011). Settlement is the valuation and transfer date; maturity is the final payment date. Accrued interest allocates the current coupon to elapsed time. Dirty price includes this accrual; clean price subtracts it (Tuckman and Serrat 2011).

2.1 Yield to maturity versus zero/spot rates versus par yields

Yield to maturity (YTM) is the single rate equating discounted promised payments to dirty price under specified compounding. A zero-coupon bond pays only at redemption: its spot, or zero-coupon, rate applies to that single horizon (Tuckman and Serrat 2011). For maturity $T > 0$ years and continuous annual decimal zero rate $r(T)$, the dimensionless discount factor is exactly

$$DF(T) = \exp[-r(T)T]. \quad (2.1)$$

A par yield is the annual coupon rate making a specified bond worth face value under given discount factors, conventionally at a coupon-date settlement with no accrual. It depends on the whole payment schedule, whereas a spot rate discounts one maturity.

The term structure describes rates across maturities, or tenors; a yield curve represents it under a stated rate convention (Tuckman and Serrat 2011).

Treasury constant-maturity (CMT) yields are read from the U.S. Treasury’s par yield curve and differ from zero rates (U.S. Department of the Treasury 2026). Fitting them does not by itself bootstrap zeros; bootstrapping infers discount factors from instrument prices and cash flows (Bank for International Settlements 2005; Tuckman and Serrat 2011). Here the invented observations have no intrinsic compounding convention; risk deliberately interprets fitted rates as continuous zeros, an educational assumption.

All rates are decimal: $0.05 = 5\%$. One basis point (bp) equals 0.01 percentage point, or 0.0001 decimal (Tuckman and Serrat 2011).

2.2 Literature and methodological position

The literature separates curve representation from estimation using instruments. Nelson–Siegel provides a compact curve family; Svensson adds flexibility through a second curvature component (Nelson and Siegel 1987; Svensson 1994). This flexibility is useful for representing shapes, but it does not establish that six parameters are necessary for these ten illustrative observations. Institutional approaches also differ: Anderson and Sleath use a spline-based method, while Gürkaynak, Sack and Wright estimate a discount function from Treasury instruments (Anderson and Sleath 2001; Gürkaynak et al. 2007). Such studies address instrument selection and estimation choices absent here. The present study assesses an existing NSS implementation; it neither compares competing curve families nor establishes empirical superiority. Its research contribution is therefore reproducible verification under specified conventions.

Chapter 3

Fixed-rate bond analytics

3.1 Cash flows and valuation

Assume fixed nonnegative coupons, positive redemption and annual or semiannual payments. Dates are maturity minus multiples of 12 or 6 months, clipping nonexistent days. Settlement-date payments are excluded.

Let $F > 0$ be face value in currency, $c \geq 0$ the annual decimal coupon rate, $m \in \{1, 2\}$ payments per year, and y_{YTM} nominal annual decimal YTM compounded m times. The dimensionless accumulation factor is $B = 1 + y_{\text{YTM}}/m > 0$. For settlement d_s , preceding/on-settlement coupon d_- , next coupon d_+ , and calendar-day count days, define accrued-period fraction α , currency coupon C , and accrued interest AI :

$$C = \frac{Fc}{m}, \quad \alpha = \frac{\text{days}(d_s - d_-)}{\text{days}(d_+ - d_-)}, \quad AI = C\alpha. \quad (3.1)$$

Index the N future payments by $i = 1, \dots, N$, with dates d_i . Their coupon-period distances $q_i = i - \alpha$ are fractional between coupon dates. Cash flow CF_i equals C , plus F at redemption; PV_i denotes its currency present value. Dirty and clean currency prices $P_{\text{dirty}}, P_{\text{clean}}$ follow exactly from fractional compounding:

$$P_{\text{dirty}} = \sum_{i=1}^N \underbrace{CF_i B^{-q_i}}_{PV_i}, \quad P_{\text{clean}} = P_{\text{dirty}} - AI. \quad (3.2)$$

Par means price equals face; premium and discount mean above and below it. At coupon-date settlement, $c = y_{\text{YTM}}$ implies $P_{\text{dirty}} = P_{\text{clean}} = F$ (section A.1); between coupons exact par is not guaranteed.

3.2 Duration, convexity and DV01

Duration measures local rate sensitivity; Macaulay duration D_{Mac} is also weighted payment time (Macaulay 1938). Define coupon-based years $T_i = q_i/m$ and abbreviate dirty price by P ; primes denote YTM derivatives. Differentiation yields modified duration D_{Mod} :

$$D_{\text{Mac}} = \frac{\sum_i T_i P V_i}{P}, \quad P' = -\sum_i C F_i \frac{q_i}{m} B^{-q_i-1}, \quad (3.3)$$

$$D_{\text{Mod}} = -\frac{P'}{P} = \frac{D_{\text{Mac}}}{B}. \quad (3.4)$$

Both durations are conventionally expressed in years (Tuckman and Serrat 2011). For a small decimal yield change δy , relative price change is approximately $-D_{\text{Mod}}\delta y$.

For larger yield changes, the approximation also uses price curvature. Normalised convexity $\mathcal{C}$, conventionally expressed in years squared (Tuckman and Serrat 2011), follows from

$$P'' = \sum_i C F_i \frac{q_i(q_i+1)}{m^2} B^{-q_i-2}, \quad \mathcal{C} = P''/P. \quad (3.5)$$

Positive redemption makes $\mathcal{C} > 0$. Taylor's local polynomial approximation gives

$$\frac{P(y_{\text{YTM}} + \delta y) - P(y_{\text{YTM}})}{P(y_{\text{YTM}})} \approx -D_{\text{Mod}}\delta y + \frac{1}{2}\mathcal{C}(\delta y)^2. \quad (3.6)$$

DV01 measures the currency price response to one basis point (Tuckman and Serrat 2011), here using down-minus-up central repricing with decimal bump $h = 10^{-4}$:

$$\text{DV01}_{\text{YTM}} = \frac{P(y_{\text{YTM}} - h) - P(y_{\text{YTM}} + h)}{2} = h P D_{\text{Mod}} + O(h^3). \quad (3.7)$$

The remainder is bounded locally by a constant times h^3 . Central repricing retains odd-order nonlinear terms beyond duration; the convexity term cancels between the symmetric prices. Both bumped yields must remain valid. Sections A.2 and A.3 provide the derivations.

Chapter 4

Nelson–Siegel–Svensson modelling

The Nelson–Siegel model combines level, slope-related and curvature contributions. Svensson’s extension adds another curvature contribution, giving the Nelson–Siegel–Svensson (NSS) model (Nelson and Siegel 1987; Svensson 1994). The parameter vector is $\theta = (\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2)$. The β coefficients are decimal rates and the positive decay scales τ_1, τ_2 are measured in years. A factor loading is a function of maturity multiplying a coefficient. For either decay scale τ , dimensionless loadings define the generic decimal rate r_{NSS} :

$$L_1(T; \tau) = \frac{1 - e^{-T/\tau}}{T/\tau}, \quad L_2(T; \tau) = L_1(T; \tau) - e^{-T/\tau}, \quad (4.1)$$

$$r_{\text{NSS}}(T; \theta) = \beta_0 + \beta_1 L_1(T; \tau_1) + \beta_2 L_2(T; \tau_1) + \beta_3 L_2(T; \tau_2). \quad (4.2)$$

Asymptotic limits, the values approached at boundaries, follow from the exponential expansions in section A.4:

$$r_{\text{NSS}}(0^+) = \beta_0 + \beta_1, \quad r_{\text{NSS}}(\infty) = \beta_0. \quad (4.3)$$

Thus β_0 sets the long limit, β_1 the short offset, and β_2, β_3 overlapping hump/trough contributions; decay scales locate the individual loadings, not necessarily turning points of their sum. These are descriptive roles, not structurally unique economic factors.

For dimensionless $x = T/\tau$, subtracting nearly equal rounded numbers in $1 - e^{-x}$ causes cancellation (Higham 2002). The implementation uses `expm1`, accurately evaluating $e^{-x} - 1$, and separate fifth-degree loading expansions below $x = 10^{-3}$.

4.1 Calibration and interpretation

Calibration selects parameters fitting observations. For n observations (u_ℓ, z_ℓ) , indexed by ℓ , maturities are years and rates decimal. Residuals e_ℓ are observed minus fitted:

$$e_\ell = z_\ell - r_{\text{NSS}}(u_\ell; \theta), \quad S(\theta) = \sum_{\ell=1}^n e_\ell^2. \quad (4.4)$$

Least squares minimises S , measured in squared decimal rates. At the selected fit $\hat{\theta}$, root mean squared error is

$$\text{RMSE} = \sqrt{S(\hat{\theta})/n}, \quad \text{RMSE}_{\text{bp}} = 10^4 \text{RMSE}. \quad (4.5)$$

Nonlinear optimisation searches for parameters minimising S ; local minima minimise it within a neighbourhood, without ensuring global optimality. Six deterministic starts reduce starting-point dependence (chapter 6). A small fitting error does not imply that parameters are uniquely identified (Gilli et al. 2010): a constant curve leaves decay scales undetermined. If $\tau_1 = \tau_2$, only the sum $\beta_2 + \beta_3$ enters their shared loading. Near equality can therefore produce unstable coefficients even when fitted rates change little. The generic rate r_{NSS} acquires zero-rate meaning only through the assumption in section 2.1.

Chapter 5

Interest-rate and curve risk

The curve-risk calculation discounts each contractual payment separately. Actual/365 times t_i count settlement-to-payment calendar days, including leap days, divided by 365. Zero-discounted present values V_i and dirty currency price $P_{\text{zero}}[r]$ are

$$t_i = \frac{\text{days}(d_i - d_s)}{365}, \quad V_i = CF_i e^{-r(t_i)t_i}, \quad P_{\text{zero}}[r] = \sum_i V_i. \quad (5.1)$$

This uses continuous compounding and the assigned zero interpretation (section 2.1), separately from coupon-based T_i and YTM pricing.

An additive decimal shock $s(T)$ gives shocked rate $r_s(T) = r(T) + s(T)$. Full repricing recalculates discounted payments; scenario profit and loss (P&L) is currency change $\Delta P_s = P_{\text{zero}}[r + s] - P_{\text{zero}}[r]$. For constant parallel amplitude δr ,

$$\frac{dP_{\text{zero}}[r + \delta r]}{d\delta r} = - \sum_i t_i V_i e^{-\delta r t_i} < 0. \quad (5.2)$$

Positive redemption ensures higher rates lower price.

Steepening increases the long-minus-short maturity rate difference; flattening decreases it. The specified steepener assigns $(-10, 0, +10)$ bp at $(2, 10, 30)$ years; the flattener reverses these shifts. Piecewise-linear interpolation joins neighbouring anchors by straight lines, holding endpoint shifts constant outside them. These definitions are illustrative, not universal.

5.1 Key-rate decomposition

Exposure varies by payment horizon (Ho 1992). Choose J ordered key maturities k_j in years, indexed by $j = 1, \dots, J$. Dimensionless basis weight $b_j(T)$ equals one at its own key and zero at others, interpolating linearly with constant endpoint tails. Neighbouring

weights sum to one, a partition of unity:

$$b_j(k_j) = 1, \quad \sum_{j=1}^J b_j(T) = 1, \quad 0 \leq b_j(T) \leq 1. \quad (5.3)$$

Key-rate DV01 measures currency sensitivity to hb_j ; parallel DV01 uses constant h :

$$\text{KRDV01}_j = \frac{P_{\text{zero}}[r - hb_j] - P_{\text{zero}}[r + hb_j]}{2}, \quad (5.4)$$

$$\text{DV01}_{\parallel} = \frac{P_{\text{zero}}[r - h] - P_{\text{zero}}[r + h]}{2}. \quad (5.5)$$

Partition makes first-order exposures additive. Independent finite bumps leave a cubic reconciliation difference $\Delta_{\text{KR}} = \sum_j \text{KRDV01}_j - \text{DV01}_{\parallel}$, derived in section A.6. Exact finite-bump equality is therefore unnecessary.

Chapter 6

Computational methodology

The implementation uses Python 3.12+. Immutable dataclasses preserve contractual inputs, while `bonds.py`, `curves.py` and `risk.py` separate nominal valuation, curve fitting and zero discounting. This makes the distinct conventions visible; callers must still supply the correct rate interpretation. NumPy supplies arrays and linear initialisation; SciPy supplies nonlinear `least_squares` (Harris et al. 2020; Virtanen et al. 2020). The trust-region reflective algorithm restricts successive local steps and respects parameter bounds (SciPy developers 2026; Branch et al. 1999; Nocedal and Wright 2006). Six fixed decay starts precede joint fitting; the successful candidate with the smallest residual sum of squares is selected (section B.3). Gilli and colleagues identify local optima and poor conditioning as separate difficulties and study a broader search heuristic (Gilli et al. 2010). These difficulties remain possible with the engine’s finite set of local starts.

Deterministic tests use fixed inputs to check identities, monotonicity, synthetic fits, invariants and boundaries. Finite differences approximate derivatives from displaced prices; central differences use symmetric displacements. Numerical tolerances specify acceptable discrepancies; truncation error comes from omitted approximation terms. Offline execution and recorded library versions support reproducibility (appendix B).

Chapter 7

Mathematical and numerical verification

The independent internal report, `paper/verification_report.tex`, supplies separately constructed cash flows, 90-digit exponential references and analytical identities. Proofs establish results under assumptions; tests provide finite numerical evidence.

Par and clean/dirty identities agree to rounding precision. Independent central repricing agrees with the engine; YTM DV01 exceeds the duration approximation by 1.93×10^{-9} currency units. Tables 7.1 and 7.2 show derivative agreement and initial second-order convergence, followed by cancellation as steps shrink.

Table 7.1: Independent normalised derivative checks. Errors are absolute differences and differences divided by the absolute analytical reference.

Quantity; step	Analytical	Numerical	Abs. error	Rel. error
$D_{\text{Mod}}; 10^{-5}$	4.2315176920	4.2315176940	1.973×10^{-9}	4.662×10^{-10}
$\mathcal{C}; 10^{-4}$	21.1104448496	21.1104452368	3.872×10^{-7}	1.834×10^{-8}

Duration and its absolute error are in years; convexity and its absolute error in years squared. Steps are decimal YTM. Source: independent verification report, numerical duration and convexity checks.

Table 7.2: Finite-difference convergence and rounding deterioration. Both normalised central formulas have $O(h^2)$ truncation error.

Decimal step h	Absolute duration error (years)	Absolute convexity error (years squared)
10^{-2}	1.951×10^{-3}	5.912×10^{-3}
10^{-3}	1.951×10^{-5}	5.911×10^{-5}
10^{-4}	1.951×10^{-7}	3.872×10^{-7}
10^{-5}	1.973×10^{-9}	1.810×10^{-6}
10^{-6}	5.499×10^{-10}	1.247×10^{-4}
10^{-7}	2.131×10^{-9}	1.832×10^{-1}
10^{-8}	2.805×10^{-8}	4.492×10^{-1}

Source: independent verification report; explicit ten-payment reference bond, rather than engine-generated reference prices.

NSS limits agree with equation (4.3); isolated loading comparisons reach maximum relative error 4.47×10^{-13} near the numerical switch. Forty independently generated noiseless observations give RMSE 3.29×10^{-18} decimal and maximum error 1.39×10^{-17} over 1,001 comparison maturities.

Independent zero-curve pricing gives 103.23571061465249 currency units, matching the engine. Shock signs satisfy equation (5.2). The basis proof covers all maturities; 100,008 sampled points have maximum computed partition error zero. Halving the reconciliation bump reduces its discrepancy approximately eightfold (section A.6). Fresh execution passes all 234 tests.

Chapter 8

Illustrative numerical results

Fresh engine calculations use the specifications in table 8.1 and illustrative term-structure observations from `examples/illustrative_curve.csv`. Table 8.2 identifies a discount bond because its clean price is below face value; its coupon is also below YTM. Modified duration implies approximately 0.423% dirty-price loss for a 10 bp rise; positive convexity moderates that loss.

Table 8.1: Illustrative bond specifications. Currency amounts refer to face value 100.

Input	YTM bond	Curve-risk bond
Settlement	15 April 2025	15 April 2025
Maturity	15 January 2030	15 January 2055
Face value	100	100
Annual coupon; frequency	4%; semiannual	4%; semiannual
Nominal annual YTM	4.5%	Not used
Discount convention	Fractional coupon periods	Continuous zeros; Actual/365

Source: author's calculations using the illustrative offline data supplied with the analytics engine.

Table 8.2: Bond analytics for the 2030 maturity under nominal YTM compounding.

Measure	Unit	Value
Accrued interest	Currency	0.994475
Dirty price	Currency	98.871313
Clean price	Currency	97.876838
Macauley duration	Years	4.326727
Modified duration	Years	4.231518
Convexity	Years squared	21.110445
YTM DV01	Currency for 1 bp	0.041838

Source: author's calculations using the illustrative offline data supplied with the analytics engine.

Tables 8.3 and 8.4 and figure 8.1 show a sub-basis-point RMSE and a trough followed by rising rates, describing only these invented observations. Parameter precision does not establish identification.

Table 8.3: NSS fitted parameters. Coefficients are decimal rates; decay scales are years.

Coefficient	Estimate	Decay scale	Estimate
β_0	0.03959735	τ_1	1.628798
β_1	0.00406525	τ_2	16.193767
β_2	-0.01904196		
β_3	0.00913973		

Source: author's calculations using the illustrative offline data supplied with the analytics engine.

Table 8.4: Calibration quality for ten illustrative observations. Residuals are observed minus fitted.

Statistic	Value
RMSE (decimal)	0.00007827624
RMSE (bp)	0.782762
Maximum absolute residual (bp)	1.520383
Successful starts	6/6
Selected termination	Status 2: <code>ftol</code>
Selected function evaluations	17

Source: author's calculations using the illustrative offline data supplied with the analytics engine.

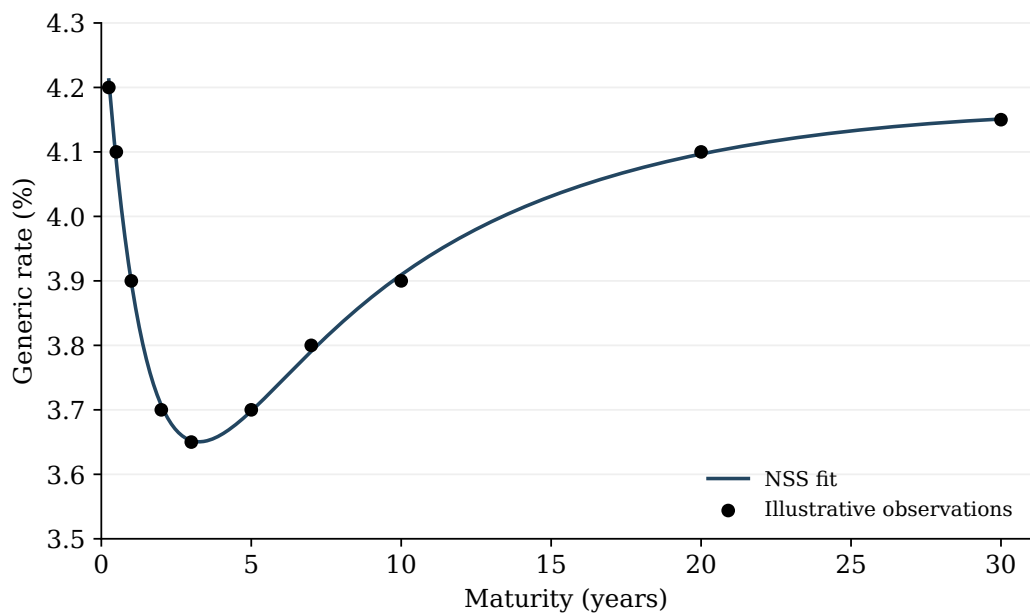


Figure 8.1: Illustrative maturity-rate observations and fitted NSS curve over the observed range. The percentage axis displays generic rates and has no implied zero-rate convention.

Source: author's calculations using the illustrative offline data supplied with the analytics engine.

Higher discount rates lower value (table 8.5); positive curve-price curvature makes the -25 bp gain larger than the $+25$ bp loss. Exposure concentrates at the 30-year key (figure 8.2 and table 8.6). The small negative reconciliation has the predicted cubic sign.

Table 8.5: Curve scenario prices and instantaneous P&L for the 2055 bond. Fitted rates are interpreted as continuous zeros.

Scenario	Base	Shocked	P&L
Parallel $+25$ bp	98.699959	94.580653	-4.119307
Parallel -25 bp	98.699959	103.072567	+4.372608
Steeper	98.699959	97.579381	-1.120578
Flattener	98.699959	99.851766	+1.151807

Currency for face 100. Shape anchors are 2/10/30 years: steeper $-10/0/+10$ bp; flattener $+10/0/-10$ bp. Cash flows and settlement remain fixed. P&L is computed before rounding, so subtracting displayed prices may differ by one last-place unit.

Source: author's calculations using the illustrative offline data supplied with the analytics engine.

Table 8.6: Key-rate DV01 and parallel reconciliation for the 2055 bond. Extra digits expose finite-bump agreement, not economic precision.

Measure	Currency for a 1 bp bump
2-year KR DV01	0.0023690644
5-year KR DV01	0.0072006030
10-year KR DV01	0.0396822358
30-year KR DV01	0.1204744674
Parallel DV01	0.1697264000
Sum of KR DV01	0.1697263707
Sum minus parallel (currency)	$-2.93434113 \times 10^{-8}$

Source: author's calculations using the illustrative offline data supplied with the analytics engine.

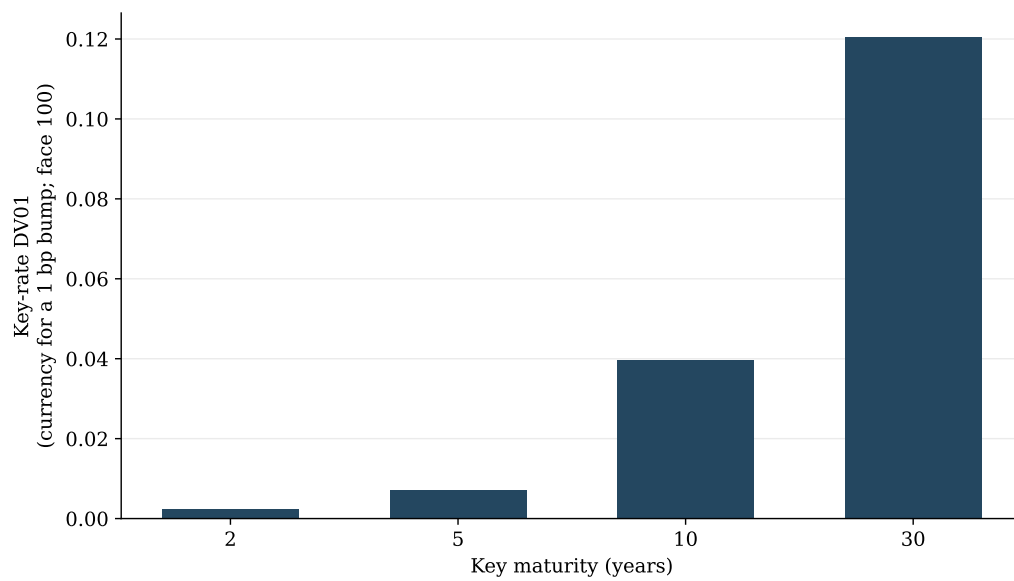


Figure 8.2: Key-rate DV01 for the illustrative 2055 bond under continuous-zero discounting and Actual/365 timing. The labelled keys are discrete categories; the vertical axis starts at zero.

Source: author's calculations using the illustrative offline data supplied with the analytics engine.

Chapter 9

Discussion and limitations

The derivations and numerical checks support the stated calculations. Model risk remains: unsuitable assumptions can produce misleading valuations.

The coupon schedule assumes regular annual or semiannual payments and excludes irregular first or final periods. There is no holiday calendar, business-day adjustment or embedded optionality; accrual and Actual/365 timing are simplified. YTM compresses maturity-dependent discounting into one rate. The assigned-zero interpretation in section 2.1 lacks instrument-based inference, unlike institutional estimation studies (Gürkaynak et al. 2007; Anderson and Sleath 2001).

The sub-basis-point RMSE measures agreement with the ten inputs used for fitting, not error on unseen observations or statistical uncertainty. Likewise, noiseless synthetic recovery checks the numerical procedure within its own model family, not that family's adequacy for market instruments. The 30-year key's large sensitivity reflects the distant redemption and late coupons, with their exposure allocated by the chosen basis.

Local calibration may miss better fits; weak identification permits substantially different parameters with similar curves. Neither arbitrage restrictions nor extrapolation validity are established. Deterministic shocks have no estimated probabilities and do not describe stochastic rate dynamics. Invented inputs support no market inference. External production-system validation and market microstructure are absent; trading or backtesting conclusions would exceed the evidence. These limits constrain economic interpretation without invalidating the conditional mathematical identities.

Chapter 10

Conclusion

The answers to the four research questions depend on the stated assumptions. Explicit coupon and compounding conventions yield coherent prices and analytical sensitivities. Stable NSS loadings and deterministic calibration represent a smooth curve, without ensuring unique parameters. Zero discounting and interpolated shocks give consistent curve exposures, with first-order key-rate additivity. Independent identities and numerical comparisons support the implementation at tested inputs. The study covers valuation, term-structure representation and deterministic risk under these assumptions; it does not establish empirical or production validity.

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Appendix A

Detailed mathematical derivations

The arguments below restate the independent verification record in the notation of table 1. Equalities are over real arithmetic unless a remainder is displayed. Cash flows, settlement and schedules remain fixed when rates change.

A.1 The par-bond identity

On a coupon date, $\alpha = 0$, $q_i = i$ and $AI = 0$. If $c = y_{\text{YTM}}$, then $C = F(B - 1)$. Thus equation (3.2) becomes

$$P = F(B - 1) \sum_{i=1}^N B^{-i} + FB^{-N}. \quad (\text{A.1})$$

For $B \neq 1$, the geometric sum telescopes:

$$\begin{aligned} (B - 1) \sum_{i=1}^N B^{-i} &= \sum_{i=1}^N (B^{1-i} - B^{-i}) \\ &= (1 - B^{-1}) + (B^{-1} - B^{-2}) + \cdots + (B^{1-N} - B^{-N}) \\ &= 1 - B^{-N}. \end{aligned}$$

Substitution into equation (A.1) gives $P = F(1 - B^{-N}) + FB^{-N} = F$. For $B = 1$, $c = y_{\text{YTM}} = 0$; coupons vanish and undiscounted redemption gives F . Clean and dirty prices coincide because accrual is zero.

More generally, at coupon-date settlement,

$$P - F = \frac{F(c - y_{\text{YTM}})}{m} \sum_{i=1}^N B^{-i}. \quad (\text{A.2})$$

Consequently coupon above YTM implies premium, and coupon below YTM implies discount, at this settlement boundary. Between coupon dates with $c = y_{\text{YTM}}$, every

present value acquires a common multiplier B^α . Hence $P_{\text{dirty}} = FB^\alpha$ and $P_{\text{clean}} = FB^\alpha - C\alpha$, which need not equal F . Fractional compounding and linear accrual are different operations.

A.2 Price derivatives and duration

Because $B = 1 + y_{\text{YTM}}/m > 0$, real powers are differentiable even for noninteger q_i . The chain rule gives

$$\frac{dB^{-q_i}}{dy_{\text{YTM}}} = -q_i B^{-q_i-1} \frac{dB}{dy_{\text{YTM}}} = -\frac{q_i}{m} B^{-q_i-1}.$$

Differentiating the finite sum term by term, using $T_i = q_i/m$ and $PV_i = CF_i B^{-q_i}$,

$$P' = -\sum_i CF_i \frac{q_i}{m} B^{-q_i-1} = -\frac{1}{B} \sum_i T_i PV_i = -P \frac{D_{\text{Mac}}}{B}. \quad (\text{A.3})$$

Dividing by $-P$ proves equation (3.4). The weights PV_i/P are nonnegative and sum to one, so Macaulay duration is a weighted mean of payment times. Positive redemption ensures $P' < 0$. Accrued interest is independent of YTM at fixed settlement; clean price has the same unnormalised derivatives, but normalising by clean price would define different sensitivities.

Differentiating again supplies another $1/m$ and changes the sign:

$$\begin{aligned} P'' &= -\sum_i CF_i \frac{q_i}{m} (-q_i - 1) B^{-q_i-2} \frac{1}{m} \\ &= \sum_i CF_i \frac{q_i(q_i + 1)}{m^2} B^{-q_i-2}. \end{aligned}$$

Normalisation gives the form evaluated by the engine:

$$\mathcal{C} = \sum_i \frac{PV_i}{P} \frac{q_i}{mB} \frac{q_i + 1}{mB} > 0. \quad (\text{A.4})$$

Both factors of $1/m$ express coupon periods in years. The factor $1/2$ belongs to Taylor's second-order price term, not the definition of convexity. Taylor's theorem at y_{YTM} , divided by P , with $P'/P = -D_{\text{Mod}}$ and $P''/P = \mathcal{C}$, yields equation (3.6) with remainder $O((\delta y)^3)$.

A.3 Central differences and DV01

Let $P^{(p)}$ denote the p -th derivative with respect to annual decimal YTM, for positive integer order p . For sufficiently small h , both perturbed yields remain inside $B > 0$.

Taylor expansion about the same yield gives

$$P(y_{\text{YTM}} \pm h) = P \pm hP' + \frac{h^2}{2}P'' \pm \frac{h^3}{6}P^{(3)} + \frac{h^4}{24}P^{(4)} + O(h^5).$$

Subtracting the upward price from the downward price cancels the even terms:

$$\begin{aligned} \frac{P(y_{\text{YTM}} - h) - P(y_{\text{YTM}} + h)}{2} &= -hP' - \frac{h^3}{6}P^{(3)} + O(h^5) \\ &= hPD_{\text{Mod}} - \frac{h^3}{6}P^{(3)} + O(h^5). \end{aligned} \tag{A.5}$$

This proves an $O(h^3)$ absolute finite-bump correction to duration-based DV01. Since

$$P^{(3)} = -\sum_i CF_i \frac{q_i(q_i + 1)(q_i + 2)}{m^3} B^{-q_i - 3} < 0,$$

the leading correction is positive. Division by two produces a currency response for the stipulated bump; division by $2h$ would instead produce a derivative per unit decimal yield.

The central derivative estimates in tables 7.1 and 7.2 obey

$$-\frac{P(y_{\text{YTM}} + h) - P(y_{\text{YTM}} - h)}{2hP} = D_{\text{Mod}} - \frac{h^2 P^{(3)}}{6P} + O(h^4), \tag{A.6}$$

$$\frac{P(y_{\text{YTM}} + h) - 2P + P(y_{\text{YTM}} - h)}{h^2 P} = \mathcal{C} + \frac{h^2 P^{(4)}}{12P} + O(h^4). \tag{A.7}$$

Each has second-order truncation error. Machine rounding introduces additional error when nearby prices are subtracted; division by a very small h , especially h^2 , amplifies it. Decreasing the step indefinitely need not improve accuracy (Higham 2002). The independent report's references are $D_{\text{Mod}} = 4.231517692026542$ and $\mathcal{C} = 21.110444849612590$; the tables preserve that record rather than treating engine output as its own reference.

A.4 NSS limits and stable loadings

Write $x = T/\tau > 0$ for either decay scale. Expanding the exponential,

$$e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} - \frac{x^5}{120} + \frac{x^6}{720} + O(x^7).$$

Subtracting from one, dividing by x , and then subtracting e^{-x} establishes

$$L_1(T; \tau) = 1 - \frac{x}{2} + \frac{x^2}{6} - \frac{x^3}{24} + \frac{x^4}{120} - \frac{x^5}{720} + O(x^6), \quad (\text{A.8})$$

$$L_2(T; \tau) = \frac{x}{2} - \frac{x^2}{3} + \frac{x^3}{8} - \frac{x^4}{30} + \frac{x^5}{144} + O(x^6). \quad (\text{A.9})$$

The constant terms give $L_1 \rightarrow 1$ and $L_2 \rightarrow 0$ as $T \downarrow 0$, proving the short limit in equation (4.3). These are limits, not evaluations at zero: the public curve interface requires positive maturity.

At large T , fixed positive decay scales give $e^{-T/\tau} \rightarrow 0$ and $0 < L_1(T; \tau) < \tau/T \rightarrow 0$. Hence $L_2 \rightarrow 0$ and $r_{\text{NSS}} \rightarrow \beta_0$. More precisely,

$$r_{\text{NSS}}(T) = \beta_0 + \frac{(\beta_1 + \beta_2)\tau_1 + \beta_3\tau_2}{T} + \text{exponentially decaying terms}. \quad (\text{A.10})$$

The short-limit offset β_1 is not the derivative at zero. The linear terms in equations (A.8) and (A.9) give

$$r'_{\text{NSS}}(0^+) = \frac{-\beta_1 + \beta_2}{2\tau_1} + \frac{\beta_3}{2\tau_2},$$

where this prime denotes the maturity derivative, in decimal rate per year, rather than a YTM price derivative.

The small- x polynomials avoid cancellation in both $1 - e^{-x}$ and $L_1 - e^{-x}$. Above the 10^{-3} switch the engine evaluates $L_1 = -\text{expm1}(-x)/x$. The fifth-degree expressions have $O(x^6)$ absolute remainders; the original exponential model remains the mathematical specification.

A.5 Key-rate basis and partition of unity

For ordered keys $0 < k_1 < \dots < k_J$, $J \geq 2$, the endpoint functions are

$$b_1(T) = \begin{cases} 1, & 0 < T \leq k_1, \\ (k_2 - T)/(k_2 - k_1), & k_1 < T < k_2, \\ 0, & T \geq k_2, \end{cases} \quad (\text{A.11})$$

$$b_J(T) = \begin{cases} 0, & T \leq k_{J-1}, \\ (T - k_{J-1})/(k_J - k_{J-1}), & k_{J-1} < T < k_J, \\ 1, & T \geq k_J. \end{cases} \quad (\text{A.12})$$

For an interior key $1 < j < J$,

$$b_j(T) = \begin{cases} (T - k_{j-1})/(k_j - k_{j-1}), & k_{j-1} < T \leq k_j, \\ (k_{j+1} - T)/(k_{j+1} - k_j), & k_j < T < k_{j+1}, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{A.13})$$

At any key only its own weight is one; other weights vanish. Below the first key only $b_1 = 1$, and above the last only $b_J = 1$. Between adjacent keys only their two basis functions are nonzero:

$$b_j(T) + b_{j+1}(T) = \frac{k_{j+1} - T}{k_{j+1} - k_j} + \frac{T - k_j}{k_{j+1} - k_j} = 1. \quad (\text{A.14})$$

These cases exhaust positive maturities and prove equation (5.3). Each weight is nonnegative and continuous, with possibly discontinuous slopes. For one key, $b_1(T) = 1$ everywhere. This proof includes the constant tails, which cannot be inferred from interpolation inside the key range alone.

A.6 First-order and finite-bump reconciliation

Let u_j be a decimal amplitude applied to basis j . With V_i fixed at the base curve,

$$P_{\text{zero}}[r + u_j b_j] = \sum_i V_i e^{-u_j t_i b_j(t_i)}.$$

Differentiating at zero and summing over keys gives

$$\sum_j \left(- \frac{\partial P_{\text{zero}}[r + u_j b_j]}{\partial u_j} \Big|_{u_j=0} \right) = \sum_i V_i t_i \sum_j b_j(t_i) = \sum_i V_i t_i. \quad (\text{A.15})$$

This equals the negative parallel derivative at zero shock in equation (5.2). Differentiating that price again gives $\sum_i t_i^2 V_i e^{-\delta r t_i} > 0$, establishing positive curvature for parallel zero-rate shifts. Multiplication by h proves exact first-order DV01 additivity.

Define bumped curves $r_j^\pm(T) = r(T) \pm h b_j(T)$ and dirty prices $P_j^\pm = P_{\text{zero}}[r_j^\pm]$. For dimensionless argument x , hyperbolic sine is $\sinh(x) = (e^x - e^{-x})/2$. Central repricing gives

$$\text{KRDV01}_j = \frac{P_j^- - P_j^+}{2} = \sum_i V_i \sinh(h t_i b_j(t_i)), \quad (\text{A.16})$$

$$\text{DV01}_\parallel = \sum_i V_i \sinh(h t_i). \quad (\text{A.17})$$

The series $\sinh(x) = x + x^3/6 + x^5/120 + \dots$ contains only odd powers. Substitution

into summed key DV01 minus parallel DV01 cancels linear terms by the partition. The resulting cubic term, denoted Δ_3 , is

$$\Delta_{\text{KR}} = \underbrace{\frac{h^3}{6} \sum_i V_i t_i^3 \left(\sum_j b_j(t_i)^3 - 1 \right)}_{\Delta_3} + O(h^5). \quad (\text{A.18})$$

For integer $p > 1$, $0 \leq b_j \leq 1$ implies $b_j^p \leq b_j$, so $\sum_j b_j^p \leq 1$. Every nonzero odd-order correction to the difference is nonpositive. Thus $\Delta_{\text{KR}} \leq 0$ in real arithmetic, strictly negative when at least one positive-value payment is split between neighbouring keys. A single key, or payments supported entirely by individual keys, gives equality.

Define hyperbolic cosine $\cosh(x) = (e^x + e^{-x})/2$. For $x \geq 0$, Taylor's theorem bounds the tail of $\sinh(x)$ after its cubic term by $x^5 \cosh(x)/120$. Each reconciliation coefficient is multiplied by $1 - \sum_j b_j^p \in [0, 1]$; the absolute omitted difference cannot exceed that positive tail. Consequently

$$|\Delta_{\text{KR}} - \Delta_3| \leq \sum_i V_i \frac{(ht_i)^5 \cosh(ht_i)}{120}. \quad (\text{A.19})$$

This bounds the analytic remainder; numerical subtraction adds rounding. For the illustrative risk bond the report records $\Delta_3 = -2.9343412800 \times 10^{-8}$ currency and bound 7.09×10^{-14} . Independent hyperbolic-sine calculations give differences $-2.34747670 \times 10^{-7}$, $-2.93434243 \times 10^{-8}$, and $-3.66792696 \times 10^{-9}$ for $h = 2 \times 10^{-4}, 10^{-4}, 5 \times 10^{-5}$, respectively. Their near-eightfold reductions confirm the cubic leading behaviour. The engine's fixed one-basis-point result in table 8.6 is consistent with this analysis.

A.7 Independent numerical constructions

The derivative reference uses explicit payment dates: 15 July 2025, each January and July in 2026–2029, and 15 January 2030. Settlement is 15 April 2025, giving $\alpha = 90/181$. Nine coupons of 2 precede redemption plus coupon of 102; $m = 2$ and $y_{\text{YTM}} = 0.045$. A separate finite-sum price function, rather than the engine's pricing method, supplies the numerical differences.

For the synthetic NSS check, the known parameter vector is

$$\theta = (0.04, -0.025, 0.03, -0.012, 1.4, 6.5),$$

with coefficient and decay units as in chapter 4. Forty geometrically spaced maturities from 0.05 to 40 years receive rates evaluated from original exponentials using 90-digit decimal arithmetic, then converted to binary64 floating point. Public calibration fits those observations; a further 1,001-point geometric grid checks interpolation. No

observation noise is added, although representation rounding remains.

The flat-zero reference uses settlement 1 April 2024, maturity 1 July 2025, coupon 6%, semiannual payments and face 100. Its explicitly specified payments are 3 on 1 July 2024, 3 on 1 January 2025 and 103 at maturity. At continuous annual zero rate 0.045, the independent value is

$$3e^{-0.045(91/365)} + 3e^{-0.045(275/365)} + 103e^{-0.045(456/365)} = 103.23571061465249$$

to displayed precision. No engine-generated schedule enters this reference.

The dense basis check takes the sorted unique union of 100,001 equally spaced maturities from 0.01 to 100 years, keys 2, 5, 10, 30, and midpoints 3.5, 7.5, 20. Independent piecewise formulas avoid calling the implementation's interpolation routine. This distinguishes independent reference construction from merely repeating the same implementation.

Appendix B

Reproducibility

B.1 Execution record

The dissertation calculations were independently rechecked on 14 September 2026 against the submitted dissertation revision 5a3f2ea. The examiner's findings and supplementary numerical evidence are recorded in `paper/examiner_report.md`. Financial source, tests and illustrative inputs are unchanged from the independent report's audited revision 7eee019. The editable installation imports this repository's package. All 234 tests passed; the individual bond, curve and risk commands completed with the results in chapter 8.

Execution used native Windows PowerShell, Python 3.12.10, NumPy 2.5.3, SciPy 1.18.1, matplotlib 3.11.2, pandas 3.0.5, python-dateutil 2.9.0.post0 and pytest 9.1.1. pandas is declared by the project but is not used in these numerical routines. Dependencies are not locked; deterministic starts ensure same-environment repeatability rather than identical results across all versions.

B.2 Environment and numerical reproduction

Run from the repository root. Create the virtual environment only when absent; installation may need network access, but the examples use only offline inputs.

```
if (-not (Test-Path .\.venv\Scripts\python.exe)) {  
    py -3.12 -m venv .venv  
}  
......
```

Check each exit code before proceeding. The curve and demo commands write `outputs/yield_curve.png`; observations are read from `examples/illustrative_curve.csv` relative to the repository root. Complete independent verification scripts remain in the standalone report and may be executed through its PowerShell stdin wrapper.

B.3 Calibration settings

Initial decay pairs, in years, are (0.5, 2), (1, 5), (2, 10), (5, 1), (10, 3), (3, 15). Each seeds linear least-squares coefficients, followed by joint six-parameter optimisation. Coefficients are unbounded; both decay scales lie in [0.01, 100] years. At least six distinct positive maturities are required; observations have equal weight, including repeated tenors.

Optimisation residuals are $10^4 e_\ell$. The installed `least_squares` defaults are trust-region reflective, linear loss and a two-point numerical Jacobian (a matrix of residual derivatives). The code specifies Jacobian-based scaling, `ftol=xtol=gtol=1e-10` and `max_nfev=3000`. These tolerances control termination through objective, step and gradient criteria; they are not bounds on financial error. Function counts exclude extra residual calls for numerical Jacobians (SciPy developers 2026).

Only successful, finite, bounded candidates qualify; smallest unscaled residual sum of squares wins, with earlier-start tie breaking. Failure of all starts raises an error. This procedure is a local search, not a guarantee of global optimality.

B.4 Figure and numerical provenance

Figure 8.1 uses the existing fitted curve evaluated at 500 evenly spaced maturities between the minimum and maximum supplied observations, with rates multiplied by 100 for percentage display. Figure 8.2 plots public key-rate DV01 results at the four default keys. Both are vector PDF figures generated using existing NumPy and matplotlib dependencies through stdin Python. The examiner corrected the key-rate figure’s clipped axis label; its regeneration script is retained in `paper/examiner_report.md`. No financial module or permanent plotting feature was added. Table values were compared with fresh command outputs or unrounded public-API results; reconciliation was computed before rounding.

B.5 LaTeX compilation

The build uses pdfLaTeX, latexmk 4.88 and Biber 2.22 with biblatex author–year references; pdfTeX is 1.40.29 (MiKTeX 26.5).

```

Push-Location .\paper
try {
    latexmk -C -outdir=build main.tex
    if ($LASTEXITCODE -ne 0) { throw "LaTeX cleanup failed." }
    latexmk -pdf -interaction=nonstopmode -halt-on-error -outdir=build main.tex
    if ($LASTEXITCODE -ne 0) { throw "LaTeX compilation failed." }
    Copy-Item .\build\main.pdf .\fixed_income_analytics_dissertation.pdf
} finally {
    Pop-Location
}
git diff --check

```

Auxiliary files remain in ignored `paper/build/`. The final PDF and two figure PDFs are intentional publication artefacts.

B.6 Bibliographic and editorial verification

All bibliography entries have verified metadata recorded in `paper/bibliography_verification.md` and are cited in the dissertation. Primary papers support the NSS attribution; institutional studies supply estimation context, with no external financial observations imported. Mathematical identities are derived rather than inferred from bibliographic authority.